

Software Test Plan (STP) - Blood Bank Management System

Project: Blood Bank Management System (BBMS)

Version: 1.0

Authors: QA Team

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Status: Finalized

1. Introduction

Purpose: This document defines the test plan for the Blood Bank Management System (BBMS) v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers BBMS features such as authentication and role-based access control, donor management, blood inventory management, hospital blood requests (including cross-matching against inventory), and blood donation drive management. The interfaces between the User Interface, business components, Database and the Notification Service are also covered. Internals of the external Notification Service and the database engine are excluded.

References: BBMS SRS v1.0, Component and Use Case diagrams, Design Specifications v1.0, IEEE 829 (Test Documentation), applicable health-data privacy guidelines.

Definitions: BBMS (Blood Bank Management System), SRS (Software Requirements Specification), RBAC (Role-Based Access Control), RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing), ABO/Rh (blood group classification), Cross-match (verifying a hospital request against compatible available stock), UI (User Interface).

2. Test Items

- User Interface component
- Authentication & RBAC module
- Donor Management module
- Inventory Management module
- Hospital Request module
- Drive Management module
- Database component (data storage and retrieval)
- Notification interface (BBMS ↔ external Notification Service)
- Admin interface

3. Features to be Tested

Features mapped to SRS requirements (BBS-F IDs shown where defined; remaining items map to SRS section 4 by feature area):

- **Authentication & RBAC:** secure login and logout, credential validation, role-based access to screens and actions (BBS-F002 for authentication-related notifications)
- **Donor Management:** donor registration, profile update, eligibility check, donation history
- **Inventory Management:** add and update blood units by blood group, stock levels, expiry tracking, expired-unit handling
- **Hospital Request:** create and track blood requests, cross-match against available inventory, fulfilment and stock update (BBS-F031)
- **Drive Management:** create and schedule donation drives, register donors, drive notifications (BBS-F042)
- **Notification Service interface:** notifications triggered correctly to the external service
- **Non-functional:** response time within SRS limit (test target ≤ 5 s), availability, data integrity (no negative stock, no double allocation of a unit), usability

4. Features Not to be Tested

- Internal logic of the external Notification Service (email/SMS delivery is assumed tested by the provider; only the interface is tested)
- Database engine internals and vendor-supplied infrastructure
- Clinical and laboratory processes (blood screening, testing, and medical eligibility decisions made outside the system)
- Network and hosting infrastructure

5. Test Approach / Strategy

Levels:

- Unit tests (module-level)
- Integration tests (UI $\leftrightarrow$ business components, business components $\leftrightarrow$ Database, Hospital Request $\leftrightarrow$ Inventory cross-match, Authentication/Drive $\leftrightarrow$ Notification stub)
- System tests (end-to-end BBMS workflows)
- Acceptance tests (UAT)

Types:

- Functional testing (core features)
- Regression testing
- Performance testing (response time, concurrent users)
- Usability testing (UI clarity, accessibility)

Entry Criteria: Stable build delivered, test data available, test environment ready.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all acceptance criteria satisfied.

5.1 Security Validation

- Validate password handling (masking, hashing, no plain-text storage or logging)
- RBAC enforcement (e.g., a hospital user cannot modify inventory or view other donors' records)
- Session timeout and account lockout after repeated failed logins
- Protection of donor personal and medical data in storage, UI and logs
- TLS 1.2+ verification (if deployed over HTTPS)
- Input validation and injection testing (SQL injection, XSS) on donor, request and search fields
- Penetration testing of authentication flows

6. Test Environment

Hardware: Client PCs/laptops with Chrome, Firefox and Edge browsers; server or cloud VM hosting the application and database; mobile browser for responsive checks.

Software: BBMS app v1.0, project database, Notification Service sandbox/stub.

Tools: Selenium (UI automation), Postman (API), JMeter (performance), Jira (defect tracking).

Test Data: Dummy donors across all ABO/Rh groups, blood units with varied expiry dates (fresh, near-expiry, expired), hospital accounts, sample requests and drives.

7. Test Schedule

Milestones:

- Test case design: 05-Oct-2026
- Environment setup: 07-Oct-2026
- Test execution start: 08-Oct-2026
- Test execution end: 17-Oct-2026
- UAT: 19-Oct-2026 to 22-Oct-2026
- Test Summary Report: 23-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Requirements Traceability Matrix (RTM)
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Bhooshan Deshpande	Prepare plan, coordinate execution
Test Engineer	Vishal H	Design & execute test cases, log defects
Developer	Aditya Umashankar	Support defect fixes and triage
Product Owner	Ullas T	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from dev team
Test environment downtime	Maintain backup environment on cloud VM
Notification Service unavailable or unreliable	Test against a stub/mock service and verify the interface contract
Time-dependent features (unit expiry, drive dates) hard to test	Seed test data with preset expiry and drive dates instead of waiting
Concurrent requests allocating the same blood unit	Add dedicated concurrency test cases and JMeter load scenarios
Ambiguous or changing requirements	Review SRS with the team early and keep the RTM updated

11. Assumptions & Dependencies

- Notification Service sandbox/stub will be stable and available
- Test data (donors, blood units, hospital accounts) will be provided before execution
- The database environment will be available and seeded before execution
- SRS requirements are frozen for the duration of the test cycle

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment unavailable for >4 hours
- Build is too unstable (blocks >30% test cases)

Resume testing if:

- Blocking defects are resolved
- Environment stabilized

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases. Example:

- Login and credential validation (Authentication) → TC-Auth-01, TC-Auth-02
- Role-based access (RBAC) → TC-Auth-03, TC-Auth-04
- Donor registration (Donor Management) → TC-Don-01, TC-Don-02
- Expiry tracking (Inventory) → TC-Inv-01, TC-Inv-02
- BBS-F031 (Hospital request cross-match) → TC-HR-01, TC-HR-02
- BBS-F042 (Drive notification) → TC-Drive-01, TC-Drive-02
- Response time (Non-functional) → TC-Perf-01

14. Test Metrics & Reporting**Metrics collected:**

- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Bhooshan Deshpande	02/10/2026
Dev Lead	Aditya Umashankar	02/10/2026
Product Owner	Ullas T	02/10/2026